

IAN SEQUEIRA

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Research Interests: ML reliability, actuarial risk modeling, algorithmic fairness, anytime-valid inference

EDUCATION

University of California, Santa Barbara

B.S. Statistics & Data Science, B.A. Economics

Santa Barbara, CA

Sept 2023 – June 2027

- GPA: 3.83 / 4.0 | Dean's Honors: 7 quarters | Graduate coursework: PSTAT 231 Statistical Machine Learning (A)
- **Selected coursework:** Design of Experiments (A+), Regression Analysis (A), Stochastic Processes (A), Data Science, Machine Learning, Econometrics (A), Probability & Mathematical Statistics, Proof Writing

RESEARCH EXPERIENCE

UCSB Department of Statistics & Applied Probability

Undergraduate Researcher (PSTAT 199) — Prof. Annie Qu

Santa Barbara, CA

Apr 2026 – Present

- Developing an anytime-valid runtime monitor for multi-agent LLM pipelines that attaches to any per-turn LLM-judge score and provides formal type-I error control at arbitrary stopping times. Targeting ICLR 2027 (submission Sept 25, 2026).
- Methodological work on e-value constructions for rubric-anchored discrete score outputs and role-conditional nulls for multi-agent settings; empirical evaluation shows nominal type-I control with meaningful detection power on held-out benchmark trajectories.

Summer Institute for Biostatistics & Data Science (iSI-BUDS)

Research Fellow — Prof. Dan Gillen & Prof. Josh Grill

UC Irvine, CA

June 2025 – Aug 2025

- Constructed GLM, ridge, and CART classifiers in R to identify Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 preclinical trial participants; best model ROC-AUC 0.835.
- Developed biomarker-guided sample-size framework achieving 80% statistical power with 188 participants vs. standard 500+.
- Contributing co-author on the resulting biomarker-guided trial design study; results appeared as a co-authored poster at CTAD 2025, San Diego (see Publications).

PUBLICATIONS AND PRESENTATIONS

Sequeira, I., Qu, A. Anytime-valid monitoring for multi-agent LLM systems. Manuscript in preparation, targeting ICLR 2027 submission (Sept 25, 2026). 2026

Gaona-Partida, P., Anuurad, T., Sequeira, I., Ragaine, E., Turner, C., Grill, J. D., Gillen, D. L. *Instructing Efficient Early-phase Preclinical Alzheimer's Disease Trials*. Manuscript in preparation. Preliminary results presented as a poster at *Clinical Trials on Alzheimer's Disease (CTAD)*, San Diego, CA, Dec 2025. 2025 – 2026

HONORS AND AWARDS

SIOP Summer Research Scholarship, UCSB Statistics & Applied Probability

(mentor: Prof. Annie Qu)

2026

Hispanic Scholarship Fund (HSF) Scholar (renewable)

2024 – Present

Dean's Honors, UC Santa Barbara (7 quarters)

2023 – 2026

PROFESSIONAL EXPERIENCE

Alida Inc.

Data Engineering Intern

Remote

Sept 2024 – July 2026

- Architected a multi-mode AI data exploration platform on Databricks operating over 47 healthcare datasets,

combining multi-agent routing, graph analysis, and automated SQL generation.

- Rewrote the platform's core join logic across nested-array datasets, cutting query complexity by 85%.
- Extended the platform with a spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training.

Turing

Applied AI/GenAI Software Engineering Intern

Remote

Feb 2026 – May 2026

- Built a 13-node LangGraph call-triage pipeline modeled after CBRE call-center operations during a 12-week ACM Industry sprint at UCSB. Pipeline classifies calls, runs a split-conformal-calibrated reranker, applies a deterministic validator layer (11 subcategory + 7 risk-override rules), gates human review, and dispatches across a 32-vendor catalog covering 52 properties.
- Composite 92.04/100 on 200-transcript dev set (10-axis scoring); zero false-911 dispatches on 800-transcript held-out test; 100% auto-resolution on 85 eligible routine cases. Chroma RAG index over 10,000 historical tickets.

TEACHING EXPERIENCE

Workshop Instructor, ACM at UCSB

2025 – 2026

- Led student workshops on applied machine learning and agentic LLM systems for undergraduate ACM members.

SELECTED PROJECTS

Healthcare Financial Toxicity Prediction

PSTAT 231 (Graduate Statistical Machine Learning) final project

Mar 2026

- Predicted catastrophic out-of-pocket burden in 20,962 MEPS 2022 respondents; tuned 6 classifiers under 10-fold cross-validation, XGBoost reaching ROC-AUC 0.874.
- Evaluated subgroup fairness across race/ethnicity using DALEX and *fairmodels*; interpreted feature contributions as SHAP odds ratios to surface actionable clinical predictors.

Research Advisor Finder

Full-stack faculty discovery platform (FastAPI, PostgreSQL/pgvector, Next.js)

Nov 2025 – Feb 2026

- Matched student research interests to 15,000+ professors across 12 STEM fields via hybrid semantic search over CV-extracted interest embeddings.

LEADERSHIP AND SERVICE

Project Executive, ACM Industry, UCSB

Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partners including Turing.

Finance & Funding Chair, CampMed, UCSB

Sept 2025 – Present

- Leading budget planning and sponsorship strategy for health-equity outreach to underrepresented high school students.

TECHNICAL SKILLS

Programming: R, Python, SQL, C++, TypeScript, LaTeX | **Methods:** e-processes, anytime-valid inference, conformal prediction, GLMs, XGBoost, CART, SHAP, DALEX, Monte Carlo | **Tools:** tidymodels, rstan, scikit-learn, PyTorch, Databricks, LangGraph

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA) | Institute of Mathematical Statistics (IMS) | UCSB Innovators Circle